

Abhirama M

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PROFESSIONAL SUMMARY

Dedicated Electronics and Communication Engineering graduate with a strong foundation in digital electronics, KiCad PCB design, and end-to-end embedded systems development. Proficient in bare-metal firmware development (ARM Cortex-M, ESP32, RP2040), direct register-level programming, and Linux system administration. Leverages practical experience from independent hardware consulting to design, assemble, and troubleshoot custom electronic solutions. Eager to apply robust component-level diagnostics and microcontroller integration skills in a professional entry-level engineering role.

EDUCATION

Bachelor of Engineering in Electronics and Communication <i>Kalpataru Institute of Technology, Tiptur</i>	2023 – 2026 CGPA: 7.56
Diploma in Electronics and Communication <i>Sanjay Memorial Polytechnic, Sagara</i>	2020 – 2023 CGPA: 9.26

TECHNICAL SKILLS

- **Hardware:** KiCad, Schematic Capture, PCB Layout, SMD/Through-Hole Soldering, Multisim, Oscilloscope, Logic Analyzer, Signal Generator, Digital Multimeter (DMM) Diagnostics, Component-Level Debugging.
- **Embedded Systems:** ARM Cortex-M (M0+/M3), RP2040, ESP32, LPC1768, FreeRTOS, I2C, SPI, UART, EEPROM, LoRa, RF, WiFi, Bluetooth, ADC/DAC.
- **Software & SysAdmin:** Embedded C, Assembly, C++, Python, MicroPython, RTOS, Linux Administration (Ubuntu Server, Mint, Manjaro), Makefiles, Docker Containerization, VirtualBox, Bash Scripting.
- **Working Knowledge:** System Verilog, Verilog, FPGA Development, Intel FPGA Starter Edition (ModelSim), PLC Programming, Human-Machine Interface (HMI), SCADA.

ACADEMIC & TECHNICAL PROJECTS

Intelligent POS System | *Embedded C, FreeRTOS, LPC1768, Keil IDE*

- Engineered a bare-metal Point of Sale (POS) system on the ARM Cortex-M based LPC1768 architecture.
- Implemented FreeRTOS for efficient task scheduling, managing concurrent processes including GPIO polling, LCD interfacing, and switch debouncing.
- Ensured strict real-time constraints and optimized dynamic memory allocation for continuous, reliable system uptime.

Universal Electronic Meter | *RP2040, MicroPython, Hardware Design*

- Designed a compact, multi-purpose measuring device utilizing a 10 mA constant current source and precise ADC voltage drop measurements for an autoranging ohmmeter circuit.
- Developed custom hardware interfaces and measurement algorithms for signal input, voltage, and current sensing using the Raspberry Pi Pico and a custom MicroPython environment.

PumpPilot (IoT Control System) | *ESP32, LoRa, Bluetooth, KiCad*

- Built an ESP32-based automatic pump control system featuring timer-based operation, non-volatile memory setting storage, and long-range LoRa communication.
- Integrated a Bluetooth module for Android app connectivity and designed the physical motor control and relay switching circuits for reliable deployment.

Linux SysAdmin & Automation | *Bash, Makefiles, Docker*

- Administered local Linux environments and provisioned VirtualBox VMs to create robust, isolated environments for firmware compilation and hardware simulation.
- Developed custom Bash and Python automation tools, integrated with Makefile build systems, to optimize C code modularization and eliminate manual repetitive compilation tasks.